

Osprey 3.5 V6 harsh shift and torque converter clutch stuck off from shift solenoid

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1 Condition

Some Osprey 3.5 V6 vehicles may exhibit harsh or delayed upshifts together with a torque converter clutch that remains stuck off. When the condition is present, DTC P0741 (Torque Converter Clutch Circuit Performance / Stuck Off) is stored in the transmission control module. Drivers may report elevated engine RPM at cruise and a lack of lockup feel. System behavior is described in the Osprey Transmission Service Manual :: Transmission System Overview .

2 Cause

This condition is caused by a sticking or electrically degraded shift solenoid pack, part number TSOL-9520, which prevents proper torque converter clutch apply. When the TCM commands lockup but the expected slip reduction is not observed, DTC P0741 is set. Contaminated fluid or internal wear within TSOL-9520 is the typical root cause. The code reference is provided in the Osprey Transmission Service Manual :: Diagnostic Trouble Code Reference .

3 Diagnostic Procedure

Confirm DTC P0741 is stored, then follow PROC-TRANS-DIAG to verify fluid level and condition before any further testing. Perform a torque converter clutch slip test and a line pressure check to isolate the fault between the solenoid pack and hydraulic circuit. Refer to the Osprey Transmission Service Manual :: Torque Converter Clutch Diagnosis and the Osprey Transmission Service Manual :: Fluid Pressure Diagnosis . The complete step sequence is given in the Osprey Transmission Service Manual :: Transmission Fault Diagnosis Procedure .

4 Repair Procedure

Replace the shift solenoid pack TSOL-9520 following PROC-SHIFT-SOL, and replace the transmission fluid and filter during the repair. After installation, clear DTC P0741, perform any required adaptive relearn, and road test to confirm smooth shifts and torque converter clutch apply. Removal and installation steps are detailed in the Osprey Transmission Service Manual :: Shift Solenoid Pack Replacement Procedure .